

AI-Powered Product Review Authenticity and Fake Review Detection System

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Abstract—Product reviews are now manipulated in two distinct ways: deceptive reviews written by paid people and reviews generated by large language models (LLMs). This paper describes a prototype review-authenticity system that models human-written deception, AI-generated text and reviewer behaviour as separate signals and combines the available signals into an explainable 0–100 Authenticity Score that reports each signal's risk score and the checks that could not be run. The two text signals use word and character TF-IDF features with logistic regression; the behaviour signal is a fully specified rule module that has not yet been evaluated quantitatively, because no labelled reviewer-history data were available. On held-out data the deception signal reached 90.9% accuracy (95% CI 87.2–93.8%) on hotel reviews, and the AI-text signal detected 91.6% of GPT-2 and 97.3% of GLM-5.2 product reviews while flagging 6.3% of genuine ones. On a small blind pilot set of 25 ChatGPT reviews, from a generator absent from training, detection rose from 2/25 for a single-generator model to 16/25 (64%, 95% CI 42.5–82.0%; exact McNemar $p = 1.2 \times 10^{-4}$). On a 9,597-review test set the fused text score reached a ROC-AUC of 0.961, and a domain gate and a length-based abstention rule, with parameters selected on validation data and evaluated offline, lowered the false-positive rate from 17.1% to 10.3% while abstaining on 18.9% of reviews. A merged single-class detector matched the text signals on machine-generated reviews but reached a ROC-AUC of only 0.853 on human-written hotel deception, against 0.972 for the dedicated model. The contribution is a capability-aware architecture and evaluation protocol rather than a new state-of-the-art accuracy figure.

Keywords—fake review detection; AI-generated text detection; deceptive opinion spam; cross-generator evaluation; multi-signal fusion; abstention; domain gating; explainable AI.

I. INTRODUCTION

Online product reviews are one of the strongest signals shoppers use to decide what to buy, which makes them a high-value target for manipulation. Two manipulation strategies now coexist. In the first, a person is paid to write a convincing but untruthful review. In the second, a large language model is prompted to generate a review that reads as genuine. Human readers and LLM-based judges have been found to be close to chance when separating real product reviews from LLM-generated ones [1], so the problem cannot be solved by inspection.

Most detection systems address only one of these phenomena or fold both into a single fake label. Detectors of machine-generated reviews are also often evaluated on text from the same generator that produced their training data, which can overstate performance on reviews from new LLMs. Finally, most systems return a verdict for every input, including very short reviews and reviews from domains on which the model was never validated.

This paper describes a prototype review-authenticity system designed around these observations and reports its evaluation. The contributions are: (1) an architecture in which

human-written deception, AI-generated text and reviewer behaviour are modelled separately and combined by coverage-gated fusion that reports each signal's risk score and never treats an unavailable signal as evidence of authenticity; (2) a cross-generator protocol in which the AI-text signal is trained on two generators and tested on a third, unseen generator, reported as a small-scale pilot; (3) an offline evaluation of two capability guards, domain gating and length-based abstention, whose parameters are selected on validation data and frozen before testing; and (4) a same-protocol comparison with conventional baselines, including a merged single-class detector. None of these techniques is claimed as individually novel; their combination was rare in the literature we surveyed. The behaviour signal is specified in full but not yet evaluated quantitatively, so the evidence reported here concerns the text signals and their fusion.

Section II reviews related work and Section III states the research gap. Section IV describes the architecture and the status of each component. Section V details the data, protocol and baselines, and Section VI reports the results. Section VII presents the prototype as qualitative evidence, and Section VIII concludes.

II. RELATED WORK

A. AI-Generated Review Detection

Meng et al. [1] report that human raters averaged 50.8% accuracy when separating real product reviews from LLM-generated ones and that LLM judges did no better; their evaluation concerns human and LLM judges rather than trained detectors, so these figures are not comparable with classifier accuracy. Salminen et al. [2] generated fake product reviews with ULMFiT and GPT-2 and showed that a machine classifier detected the GPT-2 reviews near-perfectly while human raters were markedly less accurate; their corpus supplies the GPT-2 and genuine classes used here. AlQadi et al. [3] fine-tuned DistilBERT, explained with LIME and SHAP, to detect ChatGPT-rephrased reviews and reported 97.25% accuracy on a balanced dataset built with the same process as their training data. MAiDE-up [4] provides 10,000 real and 10,000 GPT-generated hotel reviews across ten languages. Outside the review domain, the M4 study [5] reports that detectors of machine-generated text generalise poorly to unseen domains and generators, whereas Alluri et al. [6] found negligible cross-generator degradation among five instruction-tuned generator families of similar size but large losses under domain shift. Robustness to an unseen generator therefore depends on how different that generator is, which motivates testing it directly.

B. Human-Written Deception Detection

Ott et al. [7], [8] built the gold-standard corpus used by the deception signal here: truthful hotel reviews and deceptive reviews written by paid crowdworkers, on whose positive-polarity half text classifiers reached nearly 90% accuracy [7].

Later work addresses transfer across review domains. Gupta et al. [9] use orthogonal counterfactual representations for zero-shot transfer across product, hotel, restaurant and doctor reviews; Al-yousef et al. [10] propose a self-supervised transformer framework for cross-domain detection; and Cao et al. [11] apply unsupervised domain adaptation. These studies adapt to domain shift. The guard evaluated here takes the more conservative approach of withholding a signal outside its validated domain.

C. Reviewer-Behaviour-Based Detection

Behavioural cues such as posting bursts, rating anomalies and duplicated content are an established basis for detection. Ji et al. [12] use review bursts, located by kernel density estimation, to discover spammer groups without supervision. Rayana and Akoglu [13] combine linguistic, behavioural and relational cues in a review network, evaluated on labelled Yelp data, and Chen et al. [14] apply graph neural networks to identify the original spammers of astroturfing campaigns. Relational models of this kind need reviewer-product graphs and are often harder to interpret than rule-based signals.

D. Hybrid and Multi-Signal Systems

WF-CFRB [15] fuses contextual text features with reviewer-behaviour features by weighted fusion and is the closest architectural analogue to this work. Jakhar and Indian [16] propose an explainable hybrid deep learning model for fake reviews on e-commerce platforms, and Duma et al. [17] jointly model review text with overall and aspect ratings. Judging from their titles and accessible bibliographic records, none of these models machine-generated text as a separate signal. Table I summarises the key studies.

TABLE I
Summary of Key Related Studies

Study	Task and signals	Data and domain	Evaluation setting	Limitation relative to this work
Meng et al. [1]	Human and LLM-judge discrimination of real vs. LLM-generated reviews	Product reviews; three studies	Human raters (50.8% accuracy) and LLM judges	No trained detector
Salminen et al. [2]	Detection of machine-generated fake reviews	Amazon reviews; ULMFiT and GPT-2 generations	Machine classifier vs. human raters on generated data	Same generator in test; no deception or behaviour signal
AlQadi et al. [3]	ChatGPT-rephrased reviews; DistilBERT with LIME and SHAP	Constructed balanced review set	Same construction for training and test; 97.25% accuracy	Rephrasing task; no unseen-generator test
Ignat et al. [4]	GPT-generated hotel reviews	10,000 real and 10,000 generated reviews in ten languages	Detection models compared across sentiment, location and language	Hotel domain; no behaviour or fusion
Wang et al. (M4) [5]	Machine-generated text detection	Multi-generator, multi-domain, multilingual text	Unseen domains and generators	Not reviews; reports poor generalisation to unseen LLMs
Alluri et al. [6]	Human vs. machine text and code	22,141 instances (HC3, CodeSearchNet, MBPP, HumanEval) and 997 mixed documents	Cross-generator (five instruction-tuned families) and cross-domain	Not reviews; negligible cross-generator loss, large domain-shift loss
Ott et al. [7], [8]	Human-written deceptive reviews; text classifiers	1,600 hotel reviews (800 truthful, 800 crowdsourced deceptive)	Cross-validation within one corpus	One domain; no AI-text or behaviour signal
Gupta et al. [9]	Cross-domain fake reviews (COAT)	Product, hotel, Yelp restaurant, doctor and restaurant reviews	Zero-shot transfer between domains	Adapts to domain shift; no AI-text signal
Al-yousef et al. [10]; Cao et al. [11]	Cross-domain fraudulent or deceptive reviews	—	Self-supervised transformer; unsupervised domain adaptation	Adapts to domain shift; no AI-text signal†
Ji et al. [12]	Review spammer groups (GSDB)	—	Unsupervised; bursts located by kernel density estimation	Behaviour only
Rayana and Akoglu [13]	Opinion spam from linguistic, behavioural and relational cues (SpEagle)	Three Yelp review datasets	Unsupervised, optionally semi-supervised	No AI-text signal

Study	Task and signals	Data and domain	Evaluation setting	Limitation relative to this work
Chen et al. [14]	Original spammers of astroturfing campaigns; graph neural network	—	—	No AI-text signal†
Wang et al. (WF-CFRB) [15]	Weighted fusion of contextual text features and reviewer behaviours	—	—	No AI-text signal†
Jakhar and Indian [16]	Explainable hybrid deep learning for e-commerce reviews	—	—	No AI-text signal†
Duma et al. [17]	Review text with overall and aspect ratings	—	—	No AI-text or behaviour signal†

— : not stated in the abstract or bibliographic record accessible to the authors. † Judged from the title and accessible record only.

III. RESEARCH GAP AND PROBLEM STATEMENT

Across Sections II-A to II-D, several capabilities are individually established: detecting human-written deception from text, detecting behavioural anomalies, fusing text with behaviour, explaining predictions and detecting AI-generated reviews, the last usually with the same generator in training and test data.

In our targeted survey we did not find a review-authenticity system that simultaneously (i) models human-written deception and AI-generated text as separately

attributed signals, (ii) adds reviewer behaviour as a third signal, (iii) combines the available signals into an explainable score that reports which checks ran and never scores a missing check as clean, and (iv) evaluates guards that withhold a signal outside its validated domain or withhold a verdict for low-information input. Because the survey was targeted rather than systematic, this describes the surveyed literature only and is not a claim that no such system exists. The problem addressed is to build such a system, to state precisely which parts are enabled in a working prototype and which are evaluated only offline, and to measure each part with parameter selection kept separate from testing.

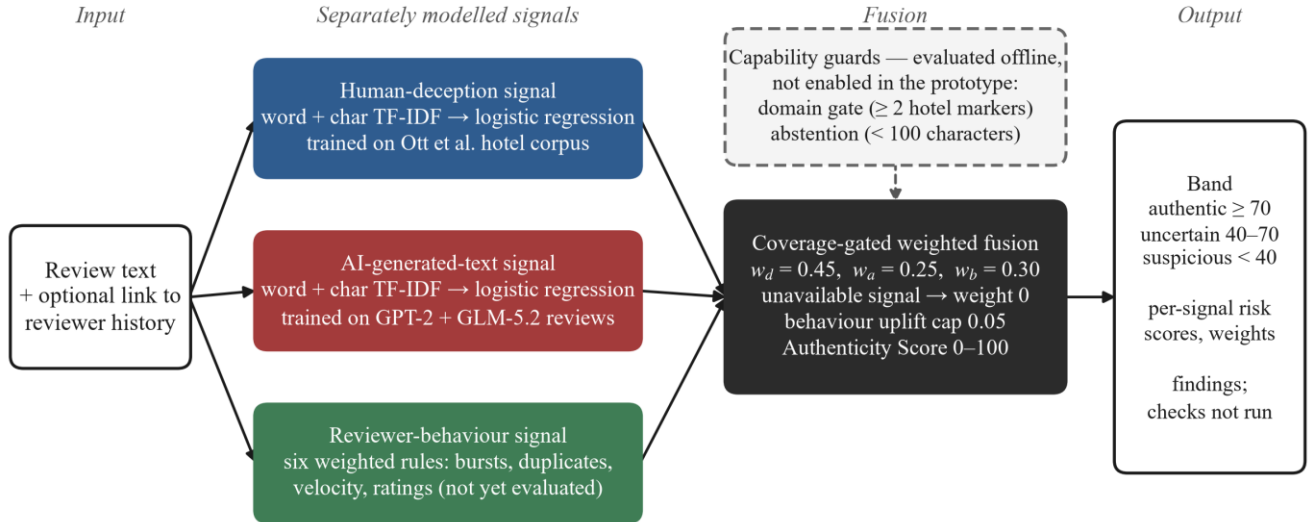


Fig. 1. Architecture. The three signals are modelled separately. Coverage-gated fusion is enabled in the prototype; the capability guards (dashed) are evaluated offline and are not enabled. The behaviour signal is specified but not quantitatively evaluated.

IV. PROPOSED SYSTEM ARCHITECTURE

A. Components and Their Status

Fig. 1 summarises the architecture. Three terms are used consistently. The *prototype* is the FastAPI and React application of Section VII, run locally for development and demonstration; it has not been deployed for public use. A component is *enabled in the prototype* if the application uses it when scoring a review: the two text signals, the behaviour rules and the coverage-gated fusion of Section IV-D. A component is *evaluated offline* if it exists only in evaluation

scripts: the domain gate and abstention rule of Section IV-E and the alternative fusion weights tested in Section VI-E.

B. Text Signals

Both text signals use the same features and classifier (Section V-C) and differ only in training data. The deception signal outputs r_d , the classifier score for the deceptive class, from a model trained on the Ott et al. corpus [7], [8]; it is validated on hotel reviews only, and its accuracy on product reviews is unmeasured. The AI-text signal outputs r_a , the score for the machine-generated class, from a model trained on GPT-2 and GLM-5.2 product reviews [2], [18]. Both

outputs lie in $[0, 1]$ but are not calibrated probabilities, so they are called risk scores throughout.

C. Reviewer-Behaviour Signal

The behaviour signal is a deterministic rule module, not a trained model. From all of a reviewer's stored reviews, with no time window, it computes the review count; the fake-review ratio ρ , the share of the reviewer's analysed reviews whose latest stored verdict is fake (a fake label from the AI-text signal when the review was analysed by the classification endpoint, or a suspicious band when it was scored by the authenticity endpoint); the posting velocity v , in reviews per active calendar day (UTC); burst days, calendar days with at least five reviews; exact duplicates, found by SHA-256 hashing of lower-cased, whitespace-collapsed text; near-duplicates, pairs of the reviewer's own reviews whose character 3-gram TF-IDF vectors have a cosine similarity of at least 0.85, ignoring texts under 20 characters; and rating statistics. The six rules in Table II add points s_k , and the reviewer risk is

$$\text{risk}_b = \min \left(100, \sum_{k=1}^6 s_k \right) \quad (1)$$

with bands low (below 40), medium (40–70) and high (70 or more). When a rule cannot be evaluated, because timestamps are missing, too few reviews have been analysed, the reviewer has too few reviews for the rating rule or more than 300 for pairwise comparison, it records a data-quality note instead of silently contributing zero. The profile is evaluable, and enters the fusion with $r_b = \text{risk}_b/100$, only if a rule fired or the reviewer has at least three reviews of which at least three have been analysed; otherwise the signal is reported as unavailable.

TABLE II
Reviewer-Behaviour Rules

Rule	Trigger	Contribution s_k
High fake-review ratio	≥ 3 analysed reviews and $\rho \geq 0.5$	$30\rho (15-30)$
Exact duplicates	≥ 1 duplicate text (SHA-256 of normalised text)	$\min(25, 10 + 5n)$
Near-duplicates	≥ 1 pair with cosine ≥ 0.85 (character 3-grams)	$\min(20, 8 + 4n)$
Posting burst	≥ 1 calendar day with ≥ 5 reviews	$\min(15, 8 + 3.5n)$
High velocity	$v \geq 3$ reviews per active day, ≥ 2 reviews	$\min(10, 5 + v)$
Extreme ratings	≥ 5 reviews, $\geq 90\%$ rated 1 or 5 stars	10

n : number of duplicates, near-duplicate pairs or burst days; ρ : fake-review ratio; v : reviews per active day. Rule weights and thresholds are engineering choices.

The rules need a reviewer identifier, review timestamps, star ratings and review texts; reviewer names and e-mail addresses are stored by the prototype but used by no rule. The fake-review-ratio rule reuses stored text verdicts, so the behaviour signal is not independent of the text signals, and a verdict stored by the authenticity endpoint can itself reflect an earlier behaviour score. Because no labelled reviewer-history dataset was available, the behaviour signal has been exercised

only on synthetic reviewer histories seeded for demonstration (Section VII-B), and its detection performance is not reported. Applying it to real platform data would require a lawful basis for processing reviewer identifiers and posting histories, pseudonymised identifiers, access control and retention limits, none of which the prototype implements.

D. Authenticity Score Formulation

Let $I = \{d, a, b\}$ index the deception, AI-text and behaviour signals, with risk scores $r_i \in [0, 1]$ (larger is riskier) and fixed weights $w_d = 0.45$, $w_a = 0.25$ and $w_b = 0.30$. For a given review, the available set $A \subseteq I$ contains each text signal whose model is loaded and the behaviour signal when the review is linked to an evaluable reviewer profile. Signals outside A receive no weight, and no score is produced if A is empty. The fused risk is the weighted mean over A :

$$R = \frac{\sum_{i \in A} w_i r_i}{\sum_{i \in A} w_i} \quad (2)$$

A clean reviewer history is weak evidence that a particular review is genuine, so when at least one text signal is available, behaviour may lower the text-only risk

$$R_T = \frac{\sum_{i \in A \setminus \{b\}} w_i r_i}{\sum_{i \in A \setminus \{b\}} w_i} \quad (3)$$

by at most $c = 0.05$, and the score is computed from the capped risk R^* :

$$R^* = \begin{cases} \max(R, R_T - c), & \text{if } b \in A \text{ and } R < R_T \\ R, & \text{otherwise} \end{cases} \quad (4)$$

$$S = \text{round}(100(1 - R^*), 1) \quad (5)$$

With both text signals available, (2)–(5) imply that a reviewer risk $r_b \geq R_T$ lowers the score by $30(r_b - R_T)$ points, whereas $r_b < R_T$ raises it by at most 5 points. Scores are mapped to the bands authentic ($S \geq 70$), uncertain ($40 \leq S < 70$) and suspicious ($S < 40$); in the evaluation, any band other than authentic counts as flagged. The weights were set heuristically when the prototype was built, informed by the component results then available, including the pilot ChatGPT detection rate of Section VI-B; they were not optimised, and Section VI-E tests them against equal and validation-selected weights. Renormalisation keeps the score defined when a signal is missing but changes its scale, so scores computed from different signal subsets are not directly comparable (Section VI-E); the prototype therefore reports the signals used with every score. Neither the score nor its bands are calibrated probabilities of fraud.

E. Capability Guards (Evaluated Offline)

Two guards were implemented in evaluation scripts but are not enabled in the prototype. The *domain gate* treats a review as inside the validated domain of the deception signal if its

lower-cased word set contains at least m distinct terms from a fixed 19-word hospitality lexicon (hotel, motel, inn, resort, suite, suites, lobby, concierge, room, rooms, checkin, housekeeping, valet, stayed, stay, nights, breakfast, amenities, guest). Otherwise the deception signal is removed from A and reported as unverified for that input, so an input whose domain is uncertain is treated as out of domain. The gate is a lexical heuristic rather than a trained classifier, and it fails in two ways that Section VI-F quantifies: product reviews that use hospitality words are misrouted, and hotel reviews with too few markers lose the deception signal. The *abstention rule* withholds the score when the stripped review is shorter than L characters and reports that the text is insufficient. Section V-B describes how m and L were selected.

V. EXPERIMENTAL SETUP

A. Datasets and Population Accounting

Table III accounts for every population. The deception signal uses the Ott et al. corpus [7], [8] of 1,600 hotel reviews,

distributed by its authors over five folds of 160 truthful and 160 deceptive reviews. The AI-text signal uses two product-review corpora: that of Salminen et al. [2], with 20,216 genuine Amazon reviews and 20,216 GPT-2 reviews, and the modern-fake-reviews-glm corpus [18], with 20,208 genuine and 20,208 GLM-5.2 reviews. The genuine halves overlap heavily: 94.8% of the genuine reviews in the second corpus are identical to reviews in the first. The 80,848 rows were therefore deduplicated by the SHA-256 hash of whitespace- and case-normalised text, leaving 61,664 unique texts, which were split 70/15/15 within each source. The per-population counts in Section VI, such as 3,029 GPT-2 and 3,191 genuine reviews, are sizes of this test split, not of the source corpora. The blind pilot set contains 25 product reviews generated with ChatGPT by the project team; they are 153–232 characters long (median 190) and are used for inference only.

TABLE III

Corpora, Splits and Evaluation Sets

Population	Domain	Source rows	Unique	Train	Set V (validation)	Set T (test)
Ott et al. truthful [7], [8]	Hotels	800	800	640 (folds 1–4)	640 (out of fold)	160 (fold 5)
Ott et al. deceptive	Hotels	800	800	640 (folds 1–4)	640 (out of fold)	160 (fold 5)
Genuine Amazon reviews [2], [18]	Products	20,216 + 20,208	21,266	14,886	3,189	3,191
GPT-2 reviews [2]	Products	20,216	20,190	14,133	3,028	3,029
GLM-5.2 reviews [18]	Products	20,208	20,208	14,145	3,031	3,032
ChatGPT pilot set (project team)	Products	25	25	—	—	25
Total		82,473	63,289	44,444	10,528	9,597

Product texts were deduplicated by SHA-256 of normalised text (80,848 $\rightarrow$ 61,664 rows) and split 70/15/15 within source over hash groups (seed 7501). Ott folds 1–4 train the deception model; in set V their deception risk scores are computed out of fold. The ChatGPT set is used for inference only.

B. Splits, Selection and Leakage Control

For the deception signal the authors' fold assignment is retained, so that reviews of the same hotel are not scattered across training and test folds; the final model is fitted on folds 1–4 and tested once on fold 5. The product corpora are split over hash groups, so no text or exact duplicate appears in two splits, and code assertions verify that no hash crosses a split boundary. Two evaluation sets are defined (Table III). The validation set V ($n = 10,528$) contains the product validation split and Ott folds 1–4, whose deception risk scores are computed out of fold by refitting the deception configuration four times, each time on three of the four folds. The test set T ($n = 9,597$) contains the product test split, Ott fold 5 and the 25 ChatGPT reviews. The fusion weight ratio, the gate parameter m and the abstention threshold L are selected on V and frozen before T is scored; T is used only for reporting.

Two methodological problems are reported rather than omitted. First, an earlier product evaluation sampled from a split generated with a different random seed from the training split, so about three-quarters of its nominally held-out samples had been seen in training; a hash-overlap audit

detected this, and all results here use the hash-grouped splits. Second, in an earlier version of this work the abstention threshold (120 characters) and the gate parameter were chosen by inspecting test populations, including the lengths and detection rates of the ChatGPT pilot set, and the fusion weights were set with the pilot detection rate in view. The ChatGPT reviews were never used to fit model parameters or the classification threshold, but they did inform those choices. In this revision m and L were re-selected on V alone, and the weights are compared with validation-selected alternatives.

C. Features, Classifiers and Reproducibility

Table IV lists the shared configuration. Each raw review is represented by a lower-cased word TF-IDF block over unigrams and bigrams and a case-sensitive character TF-IDF block over 3- to 5-grams within word boundaries. No stop-word removal, stemming or normalisation is applied, so punctuation and capitalisation remain available to the classifier, and vocabularies are selected by term frequency in the training data only. Classification uses L2-regularised logistic regression, which is convex, efficient on sparse input and has inspectable coefficients; a transformer might be more accurate but would weaken the interpretability on which the

explanations rely. For the AI-text signal, C was selected from $\{0.5, 1, 4\}$ by validation F1. For the deception signal, word-only and word-plus-character features with $C \in \{1, 4\}$ were compared by five-fold cross-validation over all folds. Because that comparison included fold 5, it was repeated with four-fold cross-validation on folds 1–4 only, which selected the same configuration (mean accuracy 0.880, against 0.876 for the next-best configuration), so the fold-5 result is not affected by the earlier overlap.

TABLE IV
Shared Features, Classifier and Reproducibility Settings

Component	Setting
Word TF-IDF	1–2-grams; 20,000 most frequent in training data; min_df 2; max_df 0.9; lower-cased
Character TF-IDF	char_wb 3–5-grams; 30,000 most frequent; min_df 3; case preserved
Weighting	sublinear TF; L2 normalisation; 50,000 features in total
Preprocessing	none (raw text)
Classifier	logistic regression, L-BFGS, L2 penalty, balanced class weights
Regularisation	$C = 4$ (AI-text: validation F1 over $\{0.5, 1, 4\}$; deception: cross-validation over word or word + char features and $C \in \{1, 4\}$)
Convergence	tolerance 10^{-4} ; cap 3,000 iterations; both models converged in 20
Decision rule	risk score ≥ 0.5 (classifier); $S < 70$ (fused score)
Random seeds	split 7501; AI-text model 7501; deception model 7502; bootstrap 20260917
Software	Python 3.13.6; scikit-learn 1.9.0 [19]; NumPy 2.5.1; SciPy 1.18.0

D. Baselines

All baselines use exactly the same splits. For each text task the proposed word-plus-character logistic regression is compared with word-only and character-only logistic regression, a word n -gram linear SVM in the style of Ott et al. [7], and multinomial naive Bayes on word n -gram counts. Hyperparameters are selected on the product validation split for the AI-text task and by four-fold cross-validation on Ott folds 1–4 for the deception task. For the end-to-end task, merged single-class detectors treat every deceptive or machine-generated review as one fake class: a word-plus-character logistic regression ($C = 4$) and a word n -gram linear SVM ($C = 1$), trained without tuning on the union of the product training split and Ott folds 1–4, because Ott folds 1–4 also belong to V . The legacy single-generator model and an intermediate model are compared on the pilot set in Section VI-B. No transformer baseline was evaluated in this revision.

E. Metrics and Statistical Analysis

Fake reviews are the positive class. Single classifiers flag a review when its risk score is at least 0.5 (a decision value of at least 0 for the SVM); fused scores flag it when $S < 70$. We report accuracy, precision, recall, F1, ROC-AUC, false-positive rate (FPR) and false-negative rate (FNR). For abstaining variants, coverage is the share of reviews that receive a score, all other metrics are computed on covered reviews, and the risk–coverage trade-off follows the reject-

option view of Chow [20]. Rates carry exact Clopper–Pearson 95% intervals [21]. Intervals for pooled metrics on T are percentile intervals from 1,000 bootstrap resamples stratified by population, and ROC-AUC intervals against the pilot set use 2,000 resamples of each class. The models compared on the pilot set score the same 25 reviews, so their detection outcomes are paired: we use the exact McNemar test [22] with the Newcombe interval for a difference of paired proportions [23], and the Wilcoxon signed-rank test [24] on the paired risk scores with the matched-pairs rank-biserial correlation as effect size. Pearson correlations carry Fisher- z intervals. Pooled metrics on T depend on its mix of 3,351 genuine and 6,246 fake reviews, only 160 of which are human-written, so per-population rates are reported alongside them.

VI. RESULTS AND DISCUSSION

A. Human-Written Deception Signal

On Ott fold 5 the deception signal reached 90.9% accuracy (95% CI 87.2–93.8%) and a ROC-AUC of 0.972, wrongly flagging 13 of 160 truthful reviews and missing 16 of 160 deceptive ones (Table V); five-fold cross-validated accuracy was 0.891 ± 0.013 . This is below the 97.25% reported for ChatGPT-rephrasing detection [3], but the two figures measure different tasks under different protocols. On the same folds the four baselines reached 90.0–90.9% accuracy and ROC-AUC 0.963–0.968 (Table VI); all accuracy intervals overlap, so the proposed configuration's advantage, visible mainly in ROC-AUC, is small.

TABLE V
Human-Deception Signal on Ott Fold 5 ($n = 320$)

Metric	Value [95% CI]
Accuracy	0.909 (291/320) [0.872, 0.938]
Precision	0.917 (144/157)
Recall	0.900 (144/160) [0.843, 0.942]
F1	0.909
ROC-AUC	0.972
FPR	0.081 (13/160) [0.044, 0.135]
FNR	0.100 (16/160) [0.058, 0.157]
Confusion matrix	[[147, 13], [16, 144]]
CV accuracy, folds 1–5	0.891 ± 0.013
CV accuracy, folds 1–4 only	0.880 (same configuration selected)

TABLE VI
Deception Baselines on Ott Fold 5 ($n = 320$)

Model (selected setting)	Accuracy [95% CI]	P	R	F1	AUC	FPR (%)
Word TF-IDF + LR ($C = 1$)	0.909 [0.872, 0.938]	0.923	0.894	0.908	0.968	7.5
Character TF-IDF + LR ($C = 4$)	0.900 [0.862, 0.931]	0.910	0.887	0.899	0.963	8.8
Word n -gram linear SVM ($C = 0.1$)	0.906 [0.869, 0.936]	0.917	0.894	0.905	0.968	8.1
Word n -gram naive Bayes ($\alpha = 1$)	0.909 [0.872, 0.938]	0.928	0.887	0.907	0.964	6.9
Word + char TF-IDF + LR (proposed) ($C = 4$)	0.909 [0.872, 0.938]	0.917	0.900	0.909	0.972	8.1

Hyperparameters selected by four-fold cross-validation on folds 1–4. P: precision; R: recall; AUC: ROC-AUC. Flagged: score ≥ 0.5 (SVM: decision value ≥ 0).

B. AI-Generated Text and Cross-Generator Generalisation

On the product test split the AI-text signal reached an accuracy of 0.942 and a ROC-AUC of 0.987 (Table VII). It detected 91.6% of GPT-2 and 97.3% of GLM-5.2 reviews, flagged 6.3% of genuine reviews and detected 16 of the 25 pilot ChatGPT reviews (64.0%, 95% CI 42.5–82.0%). With 25 reviews from one generator, this rate is imprecise and is not a general ChatGPT detection rate.

TABLE VII
AI-Text Signal by Population

Population (test split)	n	Flagged, % [95% CI]
GPT-2 (generator seen in training)	3,029	91.6 [90.6, 92.6]
GLM-5.2 (generator seen in training)	3,032	97.3 [96.7, 97.9]
ChatGPT pilot (unseen generator)	25	64.0 [42.5, 82.0]
Genuine product reviews (FPR)	3,191	6.3 [5.5, 7.2]

Product test split (n = 9,252): accuracy 0.942, precision 0.966, recall 0.945, F1 0.955, ROC-AUC 0.987. Flagged: risk score ≥ 0.5 .

Table VIII and Fig. 2 compare training configurations on the unseen generator. The legacy single-generator model, trained on GPT-2 reviews with case and punctuation removed, detected 2 of the 25 pilot reviews and separated them from genuine reviews at chance (ROC-AUC 0.505, 95% CI 0.401–0.612). Adding character features while keeping GPT-2-only data (EXP-A) did not help (1/25; ROC-AUC 0.597), whereas adding GLM-5.2 training data (EXP-B, used by the prototype) raised detection to 16/25 and ROC-AUC to 0.865 (0.775–0.936). Because all three models scored the same reviews, the comparison is paired. EXP-B detected 14 reviews missed by the legacy model and missed none that it detected (exact McNemar $p = 1.2 \times 10^{-4}$; difference 56 percentage points, 95% CI 33–72), and the rank sums of its paired score differences were 307 against 18 (Wilcoxon $p = 1.5 \times 10^{-5}$; rank-biserial $r = 0.89$). Against EXP-A the discordant counts were 15 and 0 ($p = 6.1 \times 10^{-5}$; difference 60 points, 95% CI 37–76). The earlier version of this paper applied Fisher's exact test to these counts, which ignores the pairing; the conclusion is unchanged. The legacy model was trained on a row-level random split of the GPT-2 corpus, so many of the genuine test reviews were probably in its training data; this would tend to inflate, not deflate, its ROC-AUC. The results indicate that generator diversity in the training data mattered more than the feature change, but with a 25-

review single-generator pilot they do not establish robustness to unseen generators in general.

TABLE VIII
Unseen-Generator Pilot: Paired Comparison

Model	Training data; features	Detected (of 25) [95% CI]	ROC-AUC vs. genuine [95% CI]	Paired test vs. EXP-B
Legacy	GPT-2; cleaned text; word	2 (8.0%) [1.0, 26.0]	0.505 [0.401, 0.612]	$b = 14, c = 0; p = 1.2 \times 10^{-4}$
EXP-A	GPT-2; raw text; word + char	1 (4.0%) [0.1, 20.4]	0.597 [0.500, 0.698]	$b = 15, c = 0; p = 6.1 \times 10^{-5}$
EXP-B (prototype)	GPT-2 + GLM-5.2; raw text; word + char	16 (64.0%) [42.5, 82.0]	0.865 [0.775, 0.936]	—

b: pilot reviews detected by EXP-B only; c: detected by the other model only; exact McNemar p . ROC-AUC against 3,191 genuine test reviews with 2,000 bootstrap resamples; detection intervals are Clopper–Pearson.

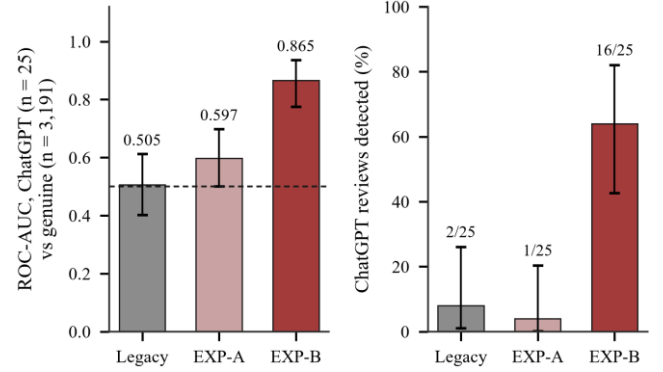


Fig. 2. Unseen-generator pilot. Left: ROC-AUC of the 25 ChatGPT reviews against 3,191 genuine test reviews (dashed line: chance). Right: ChatGPT reviews flagged at risk score ≥ 0.5 . Error bars are 95% intervals (bootstrap and Clopper–Pearson).

C. Same-Protocol Baselines

Table IX compares the AI-text signal with conventional baselines trained on the same splits. The proposed configuration has the highest accuracy (0.942), F1 (0.955) and ROC-AUC (0.987) and the lowest genuine FPR (6.3%), but its margins over word-only logistic regression and the linear SVM are small, 0.3–0.5 accuracy points. On the pilot set the five models detect 14 to 18 of the 25 reviews: naive Bayes detects the most (18) and has the highest ROC-AUC (0.884) but flags 15.2% of genuine reviews, and all pilot intervals overlap widely. Together with Table VIII, this indicates that the choice among these conventional text classifiers matters much less than the diversity of generators in the training data.

TABLE IX

AI-Text Baselines on the Product Test Split ($n = 9,252$) and the ChatGPT Pilot ($n = 25$)

Model	Selected	Accuracy	Precision	Recall	F1	ROC-AUC	Genuine FPR (%)	GPT-2 (%)	GLM-5.2 (%)	ChatGPT (of 25)	ChatGPT ROC-AUC [95% CI]
Word TF-IDF + logistic regression	$C = 4$	0.937	0.964	0.939	0.951	0.983	6.6	90.5	97.3	15	0.855 [0.758, 0.935]
Character TF-IDF + logistic regression	$C = 4$	0.919	0.953	0.922	0.937	0.974	8.6	88.1	96.2	15	0.847 [0.763, 0.919]
Word n-gram linear SVM (Ott et al. style [7])	$C = 1$	0.940	0.962	0.946	0.954	0.984	7.1	91.9	97.2	14	0.842 [0.745, 0.924]
Word n-gram multinomial naive Bayes	$\alpha = 0.1$	0.887	0.919	0.908	0.914	0.955	15.2	83.4	98.2	18	0.884 [0.809, 0.947]
Word + character TF-IDF + logistic regression (proposed)	$C = 4$	0.942	0.966	0.945	0.955	0.986	6.3	91.6	97.3	16	0.865 [0.775, 0.936]

Hyperparameters selected by F1 on the product validation split. Flagged: score ≥ 0.5 (SVM: decision value ≥ 0). ChatGPT ROC-AUC against 3,191 genuine test reviews.

D. Complementarity of the Text Signals

Table X and Fig. 3 show, for each population of T , the share of reviews that each model flags. The deception signal flags 90.0% of human-written hotel fakes but only 3.9–9.3% of machine-generated product reviews, while the AI-text signal flags 91.6–97.3% of GPT-2 and GLM-5.2 reviews and 64.0% of the pilot ChatGPT reviews but only 1.9% of human-written hotel fakes. Each signal therefore misses most of what the other detects. A linear correlation between the two risk scores does not capture this: over all of T the Pearson correlation is $r = -0.020$ (95% CI -0.040 to 0.000 ; $n = 9,597$; Spearman $\rho = 0.017$), and within populations it ranges from -0.096 to 0.057 , apart from $r = -0.35$ (CI -0.65 to 0.06) on the 25 pilot reviews. The correlation of -0.246 reported in the earlier version was computed on a smaller 1,245-review evaluation subset with a different population mix. Because a pooled correlation mostly reflects the population mix, it is no longer used as evidence, and the signals are described as separately modelled and complementary rather than independent.

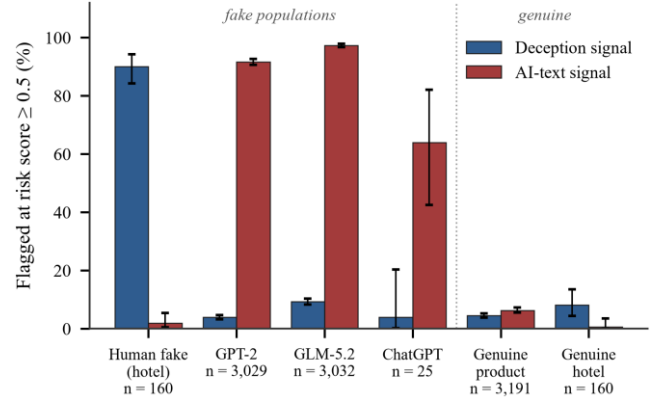


Fig. 3. Reviews flagged by each text signal (risk score ≥ 0.5) in each population of test set T , with Clopper–Pearson 95% intervals.

The benefit of separate modelling is concentrated in the minority population. A merged single-class detector with the same features and training data performs like the AI-text signal on product reviews (ROC-AUC 0.986 on the product subset of T) but reaches only 0.853 on the hotel subset, against 0.972 for the dedicated deception model, and it flags 65.0% of human-written hotel fakes and 11.9% of genuine hotel reviews (Table X). It also cannot attribute a decision to deception or to machine generation.

TABLE X

Flagged Reviews by Population on Test Set T , % [95% CI]

Population (n)	Deception, score ≥ 0.5	AI-text, score ≥ 0.5	Merged LR, score ≥ 0.5	Fusion, prototype, $S < 70$	Fusion + guards, $S < 70$ (abstained)
Human-written fake, hotel (160)	90.0 [84.3, 94.2]	1.9 [0.4, 5.4]	65.0 [57.1, 72.4]	91.9 [86.5, 95.6]	86.9 [80.6, 91.7] (0.0%)
GPT-2 (3,029)	3.9 [3.3, 4.7]	91.6 [90.6, 92.6]	91.8 [90.8, 92.8]	93.5 [92.6, 94.4]	98.0 [97.4, 98.6] (27.8%)
GLM-5.2 (3,032)	9.3 [8.3, 10.4]	97.3 [96.7, 97.9]	97.7 [97.1, 98.2]	98.4 [97.8, 98.8]	99.2 [98.8, 99.5] (14.8%)
ChatGPT pilot (25)	4.0 [0.1, 20.4]	64.0 [42.5, 82.0]	60.0 [38.7, 78.9]	72.0 [50.6, 87.9]	72.0 [50.6, 87.9] (0.0%)
Genuine product (3,191)	4.5 [3.8, 5.3]	6.3 [5.5, 7.2]	6.4 [5.5, 7.3]	17.5 [16.2, 18.8]	10.4 [9.2, 11.6] (16.5%)
Genuine hotel (160)	8.1 [4.4, 13.5]	0.6 [0.0, 3.4]	11.9 [7.3, 17.9]	10.0 [5.8, 15.7]	8.8 [4.9, 14.2] (0.0%)
ROC-AUC, hotel subset (320)	0.972	0.474	0.853	0.968	0.912
ROC-AUC, product subset (9,277)	0.526	0.986	0.986	0.961	0.992

Guards: domain gate ($m = 2$) and abstention ($L = 100$), evaluated offline; rates are on covered reviews, and the abstained share is in parentheses. ROC-AUC rows for the guarded score use covered reviews.

E. End-to-End Evaluation and Fusion Weights

Table XI compares fusion variants on T under one flagging rule. Variants that include the behaviour signal cannot be evaluated without labelled reviewer histories (Section VI-G).

The deception signal alone is not a useful detector on T (ROC-AUC 0.545), because most fake reviews in T are machine-generated. The AI-text signal alone reaches a ROC-AUC of 0.969 but, at the fused operating point, flags only 2.5% of human-written hotel fakes. The prototype's fused score

reaches a ROC-AUC of 0.961, an F1 of 0.934 and an FPR of 17.1%. On pooled metrics it is worse than the AI-text signal alone, because the hotel-trained deception signal adds a near-

constant risk of about 0.26 to product reviews, but it flags 91.9% of human-written hotel fakes.

TABLE XI
End-to-End Variants on Test Set T ($n = 9,597$)

Variant (signals used)	Coverage	Accuracy	Precision	Recall	F1 [95% CI]	ROC-AUC [95% CI]	FPR [95% CI]	FNR
Deception only	1.000	0.490	0.692	0.390	0.499 [0.486, 0.510]	0.545 [0.533, 0.556]	0.323 [0.308, 0.340]	0.610
AI-text only	1.000	0.924	0.934	0.949	0.942 [0.938, 0.945]	0.969 [0.967, 0.971]	0.124 [0.113, 0.136]	0.051
Behaviour only; AI-text + behaviour; deception + behaviour; all three signals; all three signals with guards: not evaluable (no labelled reviewer-history data)								
Deception + AI-text, prototype weights ($\lambda = 0.64$)	1.000	0.912	0.912	0.957	0.934 [0.930, 0.938]	0.961 [0.957, 0.965]	0.171 [0.158, 0.185]	0.043
Deception + AI-text, equal weights ($\lambda = 0.50$)	1.000	0.929	0.929	0.964	0.946 [0.942, 0.950]	0.975 [0.972, 0.978]	0.138 [0.126, 0.150]	0.036
Deception + AI-text, validation-selected weights ($\lambda = 0.45$)	1.000	0.930	0.932	0.963	0.947 [0.944, 0.951]	0.977 [0.974, 0.980]	0.130 [0.119, 0.142]	0.037
Prototype weights + domain gate ($m = 2$), evaluated offline	1.000	0.937	0.935	0.971	0.952 [0.949, 0.956]	0.983 [0.981, 0.985]	0.127 [0.116, 0.138]	0.029
Prototype weights + gate + abstention ($L = 100$), evaluated offline	0.811	0.951	0.944	0.982	0.962 [0.959, 0.966]	0.990 [0.988, 0.991]	0.103 [0.092, 0.114]	0.018
Merged single-class LR, $S < 70$	1.000	0.932	0.929	0.970	0.949 [0.945, 0.952]	0.983 [0.981, 0.985]	0.139 [0.128, 0.151]	0.030
Merged single-class LR, score ≥ 0.5	1.000	0.937	0.964	0.938	0.951 [0.947, 0.954]	0.983 [0.981, 0.985]	0.066 [0.058, 0.075]	0.062
Merged single-class word n-gram SVM, decision value ≥ 0	1.000	0.931	0.959	0.935	0.946 [0.942, 0.950]	0.980 [0.977, 0.982]	0.075 [0.067, 0.084]	0.065

Flagged: $S < 70$ for fused variants and single signals, unless stated. $\lambda = w_d/(w_d + w_a)$. Metrics of the abstaining variant are computed on covered reviews and are not directly comparable with the other rows. Intervals: 1,000 bootstrap resamples stratified by population.

Fusion weights. Fig. 4 plots ROC-AUC against the deception share of the text weight, $\lambda = w_d/(w_d + w_a)$. On V , ROC-AUC is highest at $\lambda = 0.45$ (0.9745, indistinguishable from $\lambda = 0.50$), against 0.964 at the prototype's $\lambda = 0.64$, and the test curve has the same shape. Equal and validation-selected weights both improve pooled ROC-AUC, F1 and FPR on T (Table XI) but lower recall on human-written hotel fakes from 91.9% to 84.4% and 74.4%, respectively. Pooled selection weights each population by its frequency, and human-written fakes are 9.6% of the fake reviews in V and 2.6% in T , so the preferred weight depends on how prevalent human deception is expected to be. The prototype's weights have not been changed; the analysis shows that they are not optimal for the pooled mix.

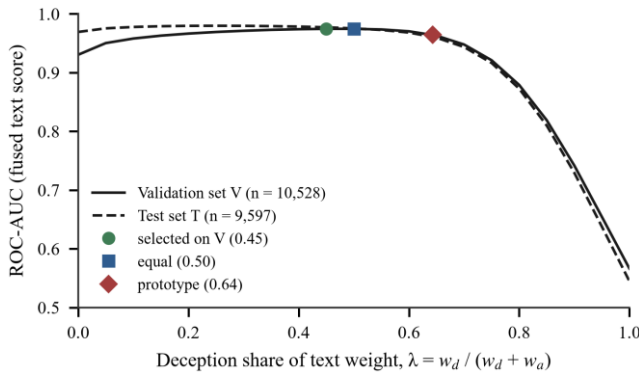


Fig. 4. Sensitivity of the fused text score to the fusion weights. Markers show the weights selected on validation set V , equal weights and the prototype's weights; the test curve is shown for transparency and was not used for selection.

Comparability across coverage levels. Renormalisation keeps the score defined when a signal is missing, but scores from different signal subsets are not on a common scale. Scoring the product reviews of T with both text signals and

with the AI-text signal alone changes the mean score of genuine reviews from 78.9 to 88.3 and that of GLM-5.2 reviews from 46.2 to 5.7, and it changes the genuine FPR from 17.5% to 12.8%. A score must therefore be interpreted together with the signals used, and band thresholds cannot be assumed to transfer between coverage levels.

F. Capability Guards (Evaluated Offline)

Domain gate. On V , $m = 2$ maximised balanced accuracy (0.975), recognising 95.3% of hotel reviews while routing 0.36% of product reviews to the hotel domain; $m = 1$ misrouted 4.6% of product reviews and $m = 3$ recognised only 77.8% of hotel reviews. On T the gate recognised 92.5% of truthful and 93.1% of deceptive hotel reviews and routed 0.3% of genuine product, 0.03% of GPT-2, 0.3% of GLM-5.2 and none of the pilot ChatGPT reviews to the hotel domain. Its 19 misrouted product reviews each contained exactly two markers, as in a clock bought for a guest room, and 22 of the 23 unrecognised hotel reviews, 11 of which were human-written fakes, contained a single marker. Gating raised product-subset ROC-AUC from 0.961 to 0.986 but lowered hotel-subset ROC-AUC from 0.968 to 0.912; over T it raised ROC-AUC to 0.983 and lowered FPR to 12.7% (Table XI). The gate is an automatic lexical check, not a domain label supplied by the platform; a platform that knows the domain of each listing could replace it with that label.

Abstention. Fig. 5 shows the risk-coverage curve of the gated score as L varies from 0 to 400 characters. A minimum validation coverage of 80% was set before the sweep, and the L that minimised validation error subject to it was selected: $L = 100$, with validation coverage 82.6% and error 5.5%. The earlier value of 120 would have reduced validation coverage to 76.2%. On T , abstaining below 100 characters covered

81.1% of reviews; on covered reviews FPR fell to 10.3% (95% CI 9.2–11.4%), F1 rose to 0.962 and ROC-AUC to 0.990 (Table XI). The rule abstained on 16.5% of genuine product, 27.8% of GPT-2 and 14.8% of GLM-5.2 reviews but on no hotel or pilot ChatGPT review, all of which exceed 100 characters. Its metrics cover only the reviews it scores and are not directly comparable with those of non-abstaining variants: the rule trades coverage for reliability.

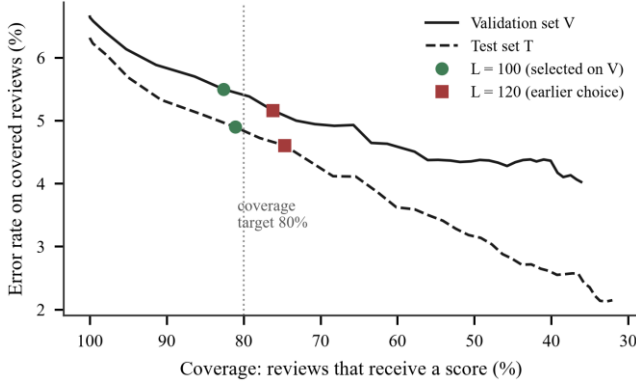


Fig. 5. Risk-coverage curves of the gated fused score as the abstention threshold L varies from 0 to 400 characters. $L = 100$ was selected on V subject to at least 80% coverage; T was not used for selection.

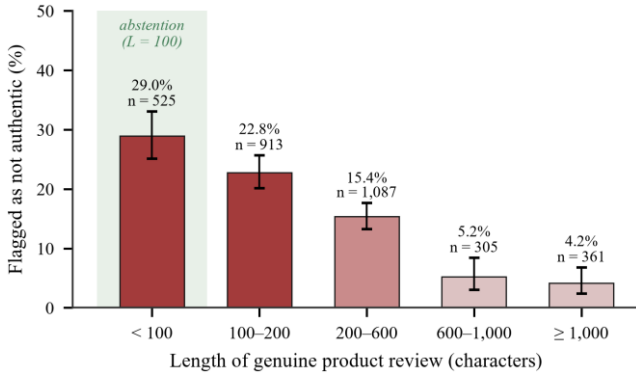


Fig. 6. Genuine product reviews in test set T flagged by the prototype's fused score ($S < 70$), by review length, with Clopper–Pearson 95% intervals. The shaded bucket falls below the abstention threshold.

Fig. 6 shows why abstention helps: the FPR of the prototype's fused score on genuine product reviews falls from 29.0% below 100 characters ($n = 525$) to 4.2% at 1,000 characters or more ($n = 361$), while their mean AI-text risk score falls from 0.212 to 0.013 and the mean deception risk score stays between 0.24 and 0.27. This is consistent with short texts carrying too little evidence for the AI-text signal.

G. Reviewer-Behaviour Signal

No labelled reviewer-history data were available, so the detection performance of the behaviour signal, and of every variant that includes it, is not reported. Its effect on the score is nevertheless exact and follows from (2)–(5). For a review whose two text risk scores are both 0.20, the score is 80.0 without reviewer history; an evaluable history in which no rule fired raises it only to 85.0 because of the uplift cap, and reviewer risks of 40, 80 and 100 lower it to 74.0, 62.0 and 56.0, the last two moving the review from the authentic to the uncertain band. Behaviour is the only signal that responds to copied or bursty posting of ordinary text, so evaluating it on labelled data with reviewer identifiers and timestamps, such as the Yelp data used by Rayana and Akoglu [13], is the main step still needed to show whether the complete three-signal system improves on its text-only form.

H. Comparison With Surveyed Systems

Table XII summarises the capabilities of the surveyed systems. In our targeted survey no system combined all of them, and we found no review system that implemented abstention or domain gating; WF-CFRB [15] is the closest analogue but does not model AI-generated text separately. The table compares capabilities, not accuracy: the systems use different data and protocols, so their reported accuracies are not comparable with ours. For the proposed system, the behaviour signal is not quantitatively evaluated and the guards are evaluated offline only.

TABLE XII
Capability Comparison With Surveyed Systems

System	Human deception	AI-generated text	Behaviour	Fusion	Explanation	Abstention	Domain gating
Ott et al. [7], [8]	✓	X	X	X	—	X	X
Salminen et al. [2]	X	✓	X	X	?	X	X
AlQadi et al. [3]	X	✓	X	X	✓	X	X
Ignat et al. (MAiDE-up) [4]	X	✓	X	X	?	X	X
Meng et al. [1]	X	✓	X	X	X	X	X
Cross-domain methods [9]–[11]	✓	X	X	X	?	X	X*
Ji et al. (burst-based) [12]	X	X	✓	X	—	X	X
Rayana and Akoglu (SpEagle) [13]	—	X	✓	—	?	X	X
Chen et al. (GNN) [14]	X	X	✓	?	?	X	X
WF-CFRB [15]	✓	X	✓	✓	?	X	X
Jakhar and Indian [16]	✓	X	?	✓	✓	X	X
Duma et al. [17]	✓	X	—	✓	?	X	X
Proposed system	✓§	✓	✓‡	✓	✓	✓†	✓†

✓ yes; X no; — partial; ? not verifiable from the accessible record. * adapts to domain shift instead of gating. † evaluated offline; not enabled in the prototype. ‡ rule-based; specified but not quantitatively evaluated. § validated on hotel reviews only.

VII. PROTOTYPE IMPLEMENTATION

The architecture is implemented as a prototype: a FastAPI service that loads both text models at start-up and exposes the fusion over a REST interface, and a React front end. The

prototype runs locally for development and demonstration, has no authentication and has not been deployed for public use. Its screenshots are qualitative evidence of how scores are explained and how missing checks are disclosed, not evidence of predictive performance, which Section VI reports.

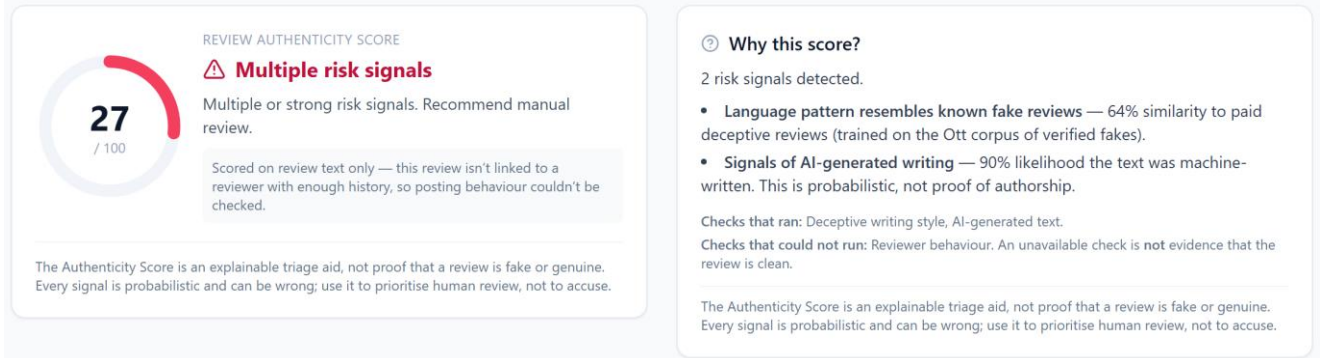


Fig. 7. Analysis output for a submitted review. The score and band (left) are paired with an explanation (right) that names each signal that fired and the checks that could not be run. The percentages shown are uncalibrated risk scores.

A. Analysis Output and Per-Signal Attribution

Fig. 7 shows the output for a submitted review. Its score of 26.7, displayed rounded to 27, falls in the suspicious band and is paired with an explanation that names each signal that fired: a deception risk score of 0.64 and an AI-text risk score of 0.90, which with (2) and (5) give the score. The interface currently renders these uncalibrated risk scores as percentages worded as similarity and likelihood, and its breakdown repeats the 64% pilot detection rate without its sample size; they should be read as risk scores and a 25-review pilot rate, and the wording will be revised. The panel lists the checks that ran and the checks that could not be run, states that an unavailable check is not evidence that the review is clean, and carries a disclaimer that the score is a triage aid rather than proof.

Fig. 8 shows the per-signal breakdown for the same review: the risk score and fusion weight of each available signal, from which the score can be recomputed, and the behaviour signal marked unavailable and excluded from (2) rather than counted as zero risk.

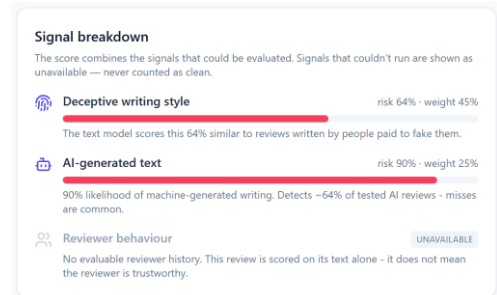


Fig. 8. Per-signal breakdown: risk score and fusion weight of each available signal; the unavailable behaviour signal is excluded, not counted as clean.

B. Reviewer-Behaviour Explanation

Fig. 9 shows the behaviour explanation for one of two synthetic reviewer histories seeded for demonstration: six five-star reviews posted at twelve-minute intervals on one day. Four of the six synthetic reviews were classified as AI-

generated, so the fake-review-ratio rule contributes 20 points; the burst (11.5, displayed as 12), velocity (10) and extreme-rating (10) rules also fire, giving a reviewer risk of 51.5, and each contribution is shown with the evidence that triggered it. The history and its texts are synthetic, so the figure illustrates the explanation mechanism only.

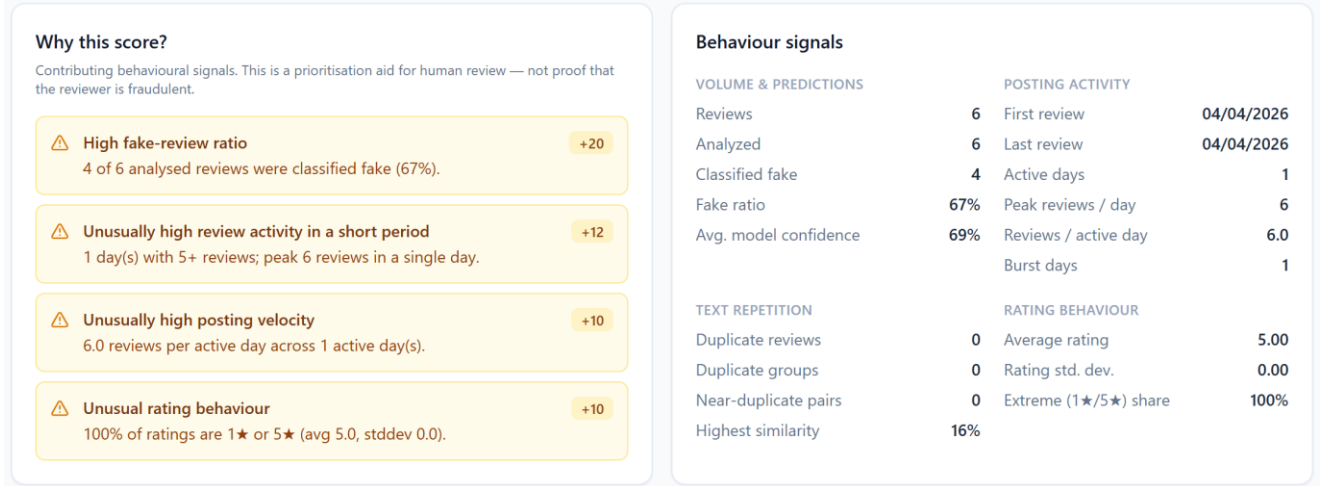


Fig. 9. Behaviour explanation for a synthetic demonstration reviewer: each fired rule is listed with its contribution and evidence (left), next to the underlying behavioural measurements (right).

C. Monitoring Dashboard

Fig. 10 shows the analytics dashboard, which aggregates stored assessments into class, confidence and time-series views. Its counts reflect development and demonstration use and are not experimental results.

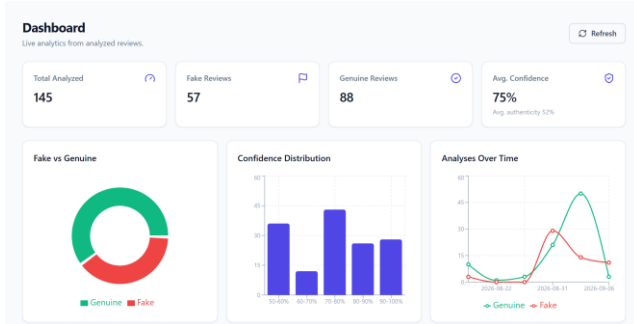


Fig. 10. Monitoring dashboard. Counts reflect development and demonstration activity, not a curated evaluation set.

VIII. CONCLUSION AND FUTURE WORK

This paper presented a prototype review-authenticity system that models human-written deception, AI-generated text and reviewer behaviour as separate signals and combines the available signals into an explainable score that reports which checks ran. The strongest evidence concerns the individual text signals: the deception signal reached 90.9% accuracy on held-out hotel reviews, and the AI-text signal detected 91.6% of GPT-2 and 97.3% of GLM-5.2 product reviews at a 6.3% false-positive rate, with a statistically

significant but small-sample gain on an unseen generator from multi-generator training (16 of 25 pilot ChatGPT reviews). For the fused text score, separate modelling preserved the detection of human-written deception that a merged detector lost, and validation-selected guards, evaluated offline, reduced false positives at a stated coverage cost. The complete three-signal system has not been evaluated end to end, because the behaviour signal lacks labelled data. The contribution is therefore a capability-aware architecture and evaluation protocol, not a new state-of-the-art accuracy figure.

Future work will evaluate the behaviour signal on labelled reviewer-history data; build a larger blind test set covering several LLMs, prompting styles and product categories; add transformer baselines under the same protocol; validate the deception signal on native product-review deception data; learn and calibrate the fusion weights; and decide from these results whether to enable the guards in the prototype.

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